

UCL EEE 2026

大二暑期融合学习计划（修订版）

数学主线 × 电磁与半导体 × 通信与光子 × 控制与模拟 × Python / MATLAB

学习周期	2026年7月27日—9月27日（9周）
每周投入	5天，每天2-3小时；总计10-15小时/周
总体目标	以ENGF0004数学工具为骨架，跨课程预习核心知识，避免重复与割裂
优先课程	ENGF0004、ELEC0019、ELEC0021；并衔接ELEC0020与ELEC0009
暂缓内容	Digital Design与Connected Systems仅做方向性准备

修订依据：UCL课程描述、个人成绩与第一周第一天诊断结果

1. 修订结论与学习策略

本计划以Mathematical Modelling and Analysis II (ENGF0004) 的数学主线为骨架, 把电磁、半导体、通信、光子、控制、模拟电子与编程挂载在共同工具上。第一周第一天安排的复数、复指数、行波方向和二阶ODE全部保留。

核心原则: 不按课程名称分别重学, 而学习五条可迁移的知识链; 每个数学工具第一次出现时学透, 随后立即用于至少两个工程领域。

五条融合知识链

知识链	共同服务的课程
复数 → Fourier → 频谱 → 波与通信	ENGF0004、ELEC0019、ELEC0020、ELEC0009、ELEC0021
ODE → Laplace → 传递函数 → Bode → 反馈	ENGF0004、ELEC0021 Control、ELEC0009、ELEC0020
向量微积分 → Maxwell → PDE → 本征模式	ENGF0004、ELEC0019、ELEC0020 Photonics
量子 → 能带 → PN/MOS → LED/激光/探测器	ELEC0019、ELEC0020、ELEC0009
矩阵 → 最小二乘 → 统计 → Python建模	ENGF0004、ELEC0020、ELEC0021及实验课程

2. 课程核对与暑期优先级

课程	标题—内容匹配	暑期优先级	建议
ENGF0004 Mathematical Modelling and Analysis II	完全匹配; 是全部课程的数学骨架	S: 最高	系统学习并用于工程问题
ELEC0019 Physics for Electronics and Nanotechnology 2	标题较宽; 实质为电磁理论+半导体器件	S/A: 很高	建立两条完整物理链
ELEC0021 Programming and Control Systems	匹配; 编程与控制为两条主线	A: 很高	控制先打基础, 编程直接进入OOP/科学计算
ELEC0020 Photonics and Communication Systems	匹配; 实质为通信+光子两部分	A/B: 高	预学Fourier、概率、波与半导体基础
ELEC0009 Analog Electronics	完全匹配	B: 中高	复习小信号、Bode、反馈与SPICE
ELEC0010 Digital Design	匹配; 偏硬件与嵌入式设计流程	C: 中低	只复习FSM、寄存器、位运算和C
ELEC0017 Connected Systems	匹配但描述宽泛; 偏IoT架构与实践	C: 低	只建立网络与IoT系统架构

3. 时间配置与任务等级

主线	标准周时间	说明
数学主线	3小时	Fourier、Laplace、向量微积分、PDE、矩阵与统计
物理主线	3小时	电磁、光子、量子与半导体
系统应用	2.5小时	Control、Communication与Analog
计算与仿真	2.5小时	Python为主, MATLAB/SPICE为辅
周测与整理	1.5小时	闭卷复现、错题、跨课程知识图

A/B/C三档任务

档位	时间	完成标准
A档：最低	约10小时	核心概念、基础题、一个融合编程任务、30分钟周复盘
B档：标准	约12.5小时	推导+基础/综合题+Python仿真+部分MATLAB/SPICE+周测
C档：挑战	约15小时	增加难题、双语言复现、拓展阅读或短技术报告

出现驾考、训练、旅行或实习时执行A档。未完成的C档任务不形成“欠账”，不得破坏下一周主线。

4. 每周五天通用节奏

学习日	重点	标准安排
第1天	数学工具	概念60–75分钟；手算45–60分钟；代码验证30分钟
第2天	物理主线	电磁/光子/半导体理论、推导与基础题
第3天	系统主线	Control/Communication/Analog应用
第4天	计算项目	Python完整程序；选关键结果用MATLAB或SPICE复现
第5天	综合与周测	统计/数据建模；闭卷周测；错题和知识图

5. 九周详细安排

第1周 | 7月27日—8月2日

复数、ODE、电场与动态系统

第1天：复数、ODE与行波

- 复数、Euler公式和复指数
- 固定相位法、行波方向、相速度/波长/频率
- 二阶常系数ODE；Python正反向波和指数解仿真

第2天：电场与电势

- Coulomb、Gauss、电势； $E=-dV/dx$
- 电子势能；Python绘制一维 $V(x)$ 和 $E(x)$

第3天：动态系统

- RC/RLC复习；自然响应、受迫响应与阻尼
- 将电路ODE连接到Control与Analog

第4天：半导体/光子起点

- 光子能量、de Broglie、bandgap
- LED与photodiode为何需要能带

第5天：编程

- 建立Wave类，储存A、omega、beta、phi
- 自动计算vp、lambda、f并绘图

本周验收

- 解释实函数、完整复函数和空间相量
- 从ODE判断振荡/衰减/阻尼
- 完成Wave类并通过两个测试案例

第2周 | 8月3日—8月9日

向量微积分、Maxwell与Poisson

数学

- scalar/vector fields；偏导
- gradient、divergence、curl、Laplacian

电磁

- Maxwell四方程；材料关系；continuity equation
- 边界条件的初步直觉

半导体

- Poisson方程；电荷—电场—电势—能带弯曲

计算

- 二维等势线、quiver场图
- 数值计算grad/div/curl，并用MATLAB复现一个图

本周验收

- 手算基础grad/div/curl
 - 解释四条Maxwell方程
 - 从电势图判断电场与电子能带方向
-

第3周 | 8月10日—8月16日

线积分、积分定理、反馈与Laplace起步

数学

- line integral、conservative field
- Gauss与Stokes定理
- Laplace定义、基本变换表

电磁

- Maxwell积分/微分形式联系；边界条件

控制/模拟

- input/output、feedback直觉
- transfer function初步；RC低通作为一阶系统

计算

- Python/MATLAB模拟一阶step response
- 比较开环与简单负反馈

本周验收

- 计算线积分和通量
 - 用积分定理解释Maxwell形式转换
 - 从一阶ODE写出基础传递函数
-

第4周 | 8月17日—8月23日

Fourier、波动方程与通信频谱

数学

- Fourier series、orthogonality
- Fourier transform、convolution

电磁

- 从Maxwell得到wave equation
- plane wave、impedance、phase/group velocity

通信

- baseband/passband; spectrum与bandwidth
- 脉冲宽度和频谱宽度的关系

计算

- 方波与脉冲FFT
- 改变脉冲宽度，验证时宽—带宽关系

本周验收

- 计算简单Fourier系数
 - 解释平面波E/H/传播方向
 - 从FFT图判断主要频率与带宽
-

第5周 | 8月24日—8月30日

Laplace、传递函数、Bode与模拟放大器

数学

- Laplace与inverse Laplace
- 用Laplace解ODE; poles和zeros

控制

- 一阶/二阶响应; damping ratio
- step response与stability直觉

模拟电子

- frequency response、cutoff、gain与phase
- Bode、small-signal与feedback

计算

- 使用SciPy或MATLAB绘制step、pole-zero和Bode图

本周验收

- 从ODE得到传递函数
 - 根据极点判断基础稳定性
 - 手绘一阶系统Bode渐近线
-

第6周 | 8月31日—9月6日

反射、折射、光纤与通信概率

电磁/光子

- reflection、Snell、全内反射
- polarization、Poynting vector、optical fibre基础

通信/统计

- Gaussian distribution、noise
- threshold detection、z-score与error probability直觉

计算

- 扫描介质阻抗比与反射系数
- 给二进制信号加入Gaussian noise并统计BER

本周验收

- 计算法向反射系数
 - 解释全内反射条件
 - 说明噪声方差如何影响BER
-

第7周 | 9月7日—9月13日

特征值、PDE、波导与量子阱

数学

- eigenvalues/eigenvectors、diagonalisation
- PDE类型、分离变量、边界值与finite difference

电磁

- TE/TM/TEM；矩形波导
- cutoff、propagation constant与modes

量子

- 定态Schrödinger；无限深势阱；tunnelling

计算

- 有限差分求量子阱能级或一维波导模式
- C档完成二者并比较模式与能级

本周验收

- 解释方程+边界条件为何产生离散本征值
 - 计算基础波导截止频率
 - 推导无限深势阱能级
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第8周 | 9月14日—9月20日

半导体器件、光电器件与模拟电子

半导体

- band diagram、Fermi level、doping
- drift/diffusion、PN、metal-semiconductor与MOS

光子

- LED、semiconductor laser、photodiode
- bandgap、发光与吸收

模拟

- BJT/MOSFET bias、transconductance
- small-signal model与单级放大器

计算/SPICE

- 绘制二极管I-V并拟合参数
- SPICE模拟简单共源或共射放大器

本周验收

- 读基础能带图
- 解释LED与photodiode的能带条件
- 画出基础小信号模型

第9周 | 9月21日—9月27日

最小二乘、假设检验、OOP与综合项目

数学/统计

- least squares、SVD直觉
- confidence interval、t/z test、ANOVA基础

编程

- class/object、composition、exceptions、testing
- NumPy/SciPy/Pandas与项目组织

综合项目

- OOP通信链路：Transmitter、Channel、Receiver、CommunicationSystem
- 生成比特、加Gaussian noise、判决、BER扫描
- 拟合结果并做基础统计；选一部分用MATLAB复现

方向性准备

- Digital: FSM、register、memory、bitwise C
- Connected: sensor→MCU→network→cloud→UI; OSI/TCP-IP概览

本周验收

- 完成综合Notebook/项目
- 形成五条知识链闭卷图
- 列出开学后5个重点追踪薄弱点

6. 工具与资源策略

主资源

- 数学：Mary L. Boas；Schey《Div, Grad, Curl, and All That》；UCL ENGF0004材料发布后以校内材料为准。
- 电磁：Sadiku或Cheng为主；MIT OCW 6.013与NPTEL作视频/习题补充。
- 半导体：Neamen为主；Chenming Hu补充MOS与CMOS scaling。
- 控制：优先学习ODE—Laplace—transfer function—Bode；使用Python SciPy与MATLAB Control工具。
- 模拟：以大一知识复习、小信号、Bode、反馈和SPICE为重点。
- 通信：先掌握Fourier、频谱、Gaussian noise和BER直觉，不提前穷尽调制公式。

计算能力目标

工具	暑期目标
Python	NumPy、Matplotlib、SciPy、Pandas；FFT、ODE、Bode、特征值、拟合、统计；OOP与测试；完整Notebook/项目。
MATLAB	矩阵、绘图、ODE、eig、FFT、transfer function与Bode；能够阅读和修改课程代码。
SPICE	建立并运行基础晶体管放大器；读取DC operating point、AC response和transient结果。

每次代码任务应提供并保存完整、独立、可运行的程序，包括imports、参数、计算、绘图与输出；并确保不依赖未定义的前序变量。

7. 暂不深入的内容

- Digital Design：复杂FPGA工具链、HDL高级语法、timing closure与ASIC physical design。
- Connected Systems：具体云平台、无线协议细节和网络安全实现。
- Control：高阶Root Locus、复杂Nyquist、完整compensator design。
- Communication：所有调制方式的完整BER推导与严格随机过程理论。
- Photonics：复杂laser rate equations、非线性光学和高级色散补偿。

8. 中断与复盘机制

- 短周（5-7小时）：只完成A档；保留数学核心、一个物理应用和一个代码任务。
- 中断一周：用1小时恢复知识图，只补关键依赖，然后直接进入当前周。
- 每周记录：实际小时、完成档位、能闭卷解释的3-5项、最多3个薄弱点、代码文件。
- 优先保留第2周向量微积分、第4周Fourier/波、第5周Laplace/Bode、第7周PDE/本征值和第8周器件链。

附录A：UCL课程官方描述

以下英文内容依据UCL课程页面整理，仅统一段落与项目符号格式。课程安排、考核方式及当年教学内容以UCL 2026/27 Module Catalogue与Moodle发布信息为准。

A1. Physics for Electronics and Nanotechnology 2 (ELEC0019)

This module is divided into two significant strands: electromagnetic theory and semiconductor devices.

Electromagnetic Theory:

This part of the course starts with a basic revision of Electromagnetics and the associated mathematical concepts and then concentrates on fundamental aspects of electromagnetic wave propagation. Starting with plane waves, these are used to introduce the basic concepts related to wave propagation such as wave velocities, polarisation, power transport, reflection and refraction. The final section of the course deals with guided waves and simple well-known waveguides. Hollow pipes are studied as a means to introduce the basic concepts, and this is followed with a short section on dielectric waveguides which include the basic structures in optical waveguiding.

The module aims to provide a fundamental understanding of electromagnetic waves and their properties in free space and in waveguides in order to enable the analysis and design of high frequency electronic systems and components.

Upon completion of this module students should be able to understand and apply the basic scientific principles underlying electromagnetic wave propagation in free space and waveguides to a range of practical situations.

Semiconductor Devices:

The second part of the course will cover fundamentals of semiconductor materials and device operation and basics of quantum mechanics. Lectures will provide basic concepts of these topics as well as problem solving to be able to apply them to more real device problems.

This course extends the insight into the solid state fundamentals and device concepts established in First Year and considers the principles of state-of-the-art silicon diode and transistor operation. A range of devices are studied including current industry standards for fabrication, including the impact of nanotechnology on current and future devices. Basics of Quantum mechanics are also taught, which covers the wave-particle duality, time-independent Schrodinger equation and electron tunnelling.

Upon the completion of the course, students will be able to:

- understand and read the semiconductor (energy-wavevector) band diagrams;
- describe basic concepts of solid state semiconductor physics;
- solve time-independent Schrodinger equations for simple cases;
- describe formation, electrostatics and transport of metal-semiconductor contacts;
- describe formation, electrostatics and transport of metal-oxide-semiconductor contacts;
- understand the scaling of CMOS technologies.

A2. Photonics and Communication Systems (ELEC0020)

This course is divided into two parts, Communication Systems and Photonics.

Communication Systems offers engineering students a grounding in the principles of digital communications systems, covering both the theory and practice of communications systems and schemes. This element aims to provide an introduction to digital communications systems, including: Baseband digital transmission systems, baseband digital reception systems, passband digital transmission systems, passband digital reception systems, and examples of how these concepts can be applied to practical communication systems.

By the end of the module students should:

- Be familiar with the basic elements of a digital communications systems
- Be familiar with the basic mathematical tools in signals and systems analysis
- Consider different baseband and passband digital transmission schemes, their advantages and disadvantages
- Understand how to calculate the spectrum and bandwidth of different baseband and passband digital transmission schemes
- Consider different baseband and passband digital reception schemes, their advantages and disadvantages
- Understand how to calculate the error probability associated with different digital reception schemes
- Be able to carry out basic analysis by applying these principles to design real-world communication systems

The Photonics section of the course introduces engineering students to the interaction of optical signals with materials, and to the design of photonic devices and systems, including optical fibre communications systems. This includes the

interaction of light / optical signals with materials, the generation and detection of light, the principles of the guiding of light, the design of photonic devices and systems, and optical communication systems.

By the end of the module students should:

- Be familiar with the range of applications of photonics and its role in industry and everyday life
- Understand different representations and models of light
- Develop a basic understanding of how light interacts with materials
- Understand how light can be generated in semiconductors, including an appreciation of the relevant properties of semiconductor materials
- Understand the principles and be able to perform basic analysis of lasers
- Understand the structure and characteristics of light-emitting diodes and semiconductor lasers
- Understand how p-n junction diodes can be used as photodetectors
- Carry out basic analysis of photodiodes
- Have an appreciation of optical waveguides and different kinds of optical fibres
- Understand the key elements of an optical communication system
- Understand the factors that limit the performance of optical communications systems
- Be able to calculate the performance of simple optical communications systems

A3. Mathematical Modelling and Analysis II (ENGF0004)

Mathematical Modelling and Analysis II builds from the content and learning in ENGF0003 Mathematical Modelling and Analysis I and helps you to further develop your understanding of the core mathematical concepts, and modelling and analysis skills that underpin the practice of engineering. This module will help you to develop the mathematical tools and approaches you will need throughout your career, regardless of your specialism.

The mode of delivery for this module is a blended learning approach integrating video recorded lecture material, live weekly workshops and biweekly Question & Answer lectures, structured online Moodle activities and resources, and computer programming and simulation using mathematical tools like MATLAB. Within the workshops, you will work on set problems that reinforce taught concepts and provide an opportunity to introduce how mathematics is applied in an engineering context. As this module follows on from ENGF0003, its syllabus includes more advanced mathematical concepts in the areas of:

Series and Transforms

- Fourier series and transforms
- Laplace series and transforms
- Using Laplace transforms to solve ODEs

Partial Differential Equations

- Types of Partial Differential Equations and initial value problems
- Solving PDEs using separation of variables
- Numerical solutions to PDEs using finite difference approximations

Vector Calculus

- Vector fields and scalar fields
- The Del operator, gradient, divergence and curl
- Line integrals, conservative forces
- Double and triple integrals

Matrices

- Solving systems of linear equations using Gaussian elimination
- Eigenvalues & eigenvectors of matrices with applications
- Matrix decompositions with applications: diagonalisation, LU factorisation, and singular value decomposition
- Least squares fitting

Statistics and Data Modelling

- Hypothesis testing
- t-testing, Z-value testing
- ANOVA (analysis of variance)

Intended Learning Outcomes

At the end of this Module, providing you engage with all of the learning activities, you should be able to:

- understand and consider the connections between mathematics and engineering, and how mathematical ideas are embedded in engineering contexts
- analyse and evaluate real-world engineering systems in a mathematical framework
- understand and apply a range of advanced mathematical concepts, including Vector Calculus, Matrices and

Eigenvectors/Values, Statistics, Series and Transforms, and Partial Differential equations (PDEs) to analyse specific problems

- identify the appropriate mathematics to study an engineering problem and find a solution
- create appropriate computer programming, modelling techniques and statistical analysis to efficiently solve and evaluate the performance of engineering systems
- analyse and evaluate the behaviour of the output of mathematical models to understand the underlying physical or conceptual models of interest
- apply mathematical and engineering problem solving independently
- generate and assess mathematical results in effective and appropriate ways to varied audiences, including non-mathematical engineering audiences.

A4. Programming and Control Systems (ELEC0021)

The Programming part of this module aims to extend the students' programming skills and introduce them to advanced structured and object-oriented design concepts, providing the foundations for design and programming in any other environment.

On completion of this module, students will be able to:

- Use practical object-oriented Python programming skills to investigate and solve complex problems.
- Select and apply appropriate programming instructions, libraries, data structures and algorithms to model and solve complex problems using robust object-oriented code written in Python.
- Apply knowledge of mathematics, statistics and computer science principles to the solution of complex problems using Python libraries for maths, data science and machine learning.

The Control part of this module represents an introduction to control systems and to the analysis of the behaviour of dynamical systems. This module provides a fundamental understanding of feedback control systems and dynamical systems in terms of steady-state response and stability; it enables the study of feedback control systems with Laplace, Nyquist and Bode plots.

On completion of this module, students will be able to:

- Comprehensively know the basic principles underlying control theory; know and distinguish the mathematical and computer models relevant to transfer function analysis and be able to apply them to real-world models for control systems.
- Know and explain the behaviour of linear systems.
- Know and apply stability criteria related to control systems based on the following diagrams and methods: Root Locus, Bode, Routh Hurwitz and Nyquist stability criteria; Design compensators.
- Know and apply mathematical modelling techniques pertinent to control systems.

A5. Analog Electronics (ELEC0009)

This module builds on the concepts in analog electronics that students have encountered in their first year, including fundamentals of low and high frequency small signal analysis of BJT and MOSFET circuits. Fundamental concepts such as filters, feedback, bode plots, power amplifiers and oscillators will be explored. The aim of this module is to give students a good introduction to the analysis of standard circuit configurations, such as single-stage amplifiers and feedback circuits, and the design of advanced circuits, such as operational amplifiers and oscillators.

By the end of the module students should be able to:

- Understand basic single-transistor amplifier configurations, i.e., CE, CB and emitter follower for BJT, and CS, CG and source follower for MOSFET.
- Use small-signal equivalent circuits in analysing circuit performance
- Analyse the low-frequency response of discrete-circuit amplifiers and estimate the corner frequency.
- Analyse the high-frequency response of discrete and integrated amplifiers using Miller theorem and open-circuit time constant methods
- Sketch the magnitude and phase response of an amplifier in a Bode plot
- Understand filter basics
- Understand general structure of negative feedback amplifiers and the basic principle that underlies their operation
- Use positive or negative feedbacks in realizing analog circuit blocks
- Design differential and multi-stage amplifiers down to transistor level
- Understand the different classes of power amplifier and be able to analyse them (class A, B, and AB)
- Learn how positive feedback is applied to make oscillators and be able to analyse different types of oscillator circuits
- Understand the operating principles of tuned amplifiers

- Use SPICE to simulate circuit behaviour

A6. Digital Design (ELEC0010)

This module introduces students to the design process for hardware-based digital solutions such as FPGAs and ASICs and software-based embedded micro-controller solutions.

On completion of this course, students should be able to:

- Understand the differences between microcontrollers, microprocessors and programmable gate arrays.
- Capture the application requirements and develop it into a system design from a functional point of view.
- Break the system down into the smallest possible implementable and verifiable subsystems.
- Develop algorithms or circuits to implement the subsystems.
- Choose the appropriate digital platform for physical implementation.
- Choose the appropriate approach to optimize for speed and area to meet the system requirements.
- Implement the entire design process from specification through to realization in digital hardware and testing.
- Identify and explain the features and functional blocks of a microcontroller.
- Describe the architecture of a microcontroller and the functionality of memory.
- Distinguish between system components and microcontroller peripherals.
- Apply logic operations to manipulate registers.
- Explain the structure of a C program for microcontrollers.

A7. Connected Systems (ELEC0017)

The modern world is full of connected systems. They underpin the information revolution and drive advances from the way we do business to the way we deliver healthcare.

This module will introduce students to connected systems by providing a comprehensive coverage of the systems and technologies that build modern communications networks. Specifically, it covers the state-of-the-art technologies of the internet of things as well as the applications that are enabled by it.

On completion of this course, students should be able to:

- Identify the elements of a connected system architecture.
- Make suitable choices of technologies for different applications.
- Produce a small connected system using high level tools.

附录B：每周复盘模板

复盘项	填写内容
完成档位与时间	A / B / C；实际小时数
能闭卷解释	3-5个概念、推导或工程联系
仍然模糊	最多3项
程序成果	文件名、图表、Notebook、MATLAB或SPICE结果
主要错误	符号、单位、边界条件、数学操作、代码或物理解释
下周调整	保留、缩减或增加的任务

最终目标：不是在九周内“学完大二”，而是形成五条可以反复调用的共同主线，使开学后能够识别数学结构、物理模型、系统表示和计算方法。